

Research Management System - Detailed Software Requirements Specification

1. Introduction

This document defines a comprehensive research management platform for hospitals, universities, research institutes, and ethics committees. The system manages research projects from proposal submission through publication.

2. Objectives

Provide proposal review, automated M-Pesa payment processing, reviewer workflows, ethics approval, progress monitoring, final paper review, publication management, certificate generation, and audit tracking.

3. User Roles

Researcher, Reviewer, Committee Chairperson, Research Administrator, Finance Administrator, Super Administrator.

4. Research Lifecycle

Draft Proposal → M-Pesa Payment → Proposal Review → Approval & Clearance Certificate → Progress Submission → Progress Review → Final Paper Submission → Final Review → Publication Approval → Completion Certificate → Archive.

5. Stage 1 - Research Proposal Submission

Researchers submit: Research Title, Abstract, Problem Statement, Background, Objectives, Research Questions, Hypotheses, Literature Review Summary, Expected Outcomes, Study Duration, Study Sites, Principal Investigator, Co-Investigators, Funding Source, Ethics Information, and Proposal Documents. Uploads: Proposal PDF, Budget, Timeline, Ethics Forms, Supporting Documents. Validation: Required fields, file type validation, duplicate title detection, plagiarism checks. Statuses: Draft, Awaiting Payment, Submitted, Under Review, Revision Requested, Approved, Rejected.

6. Proposal Review Workflow

Admin assigns reviewers. Reviewers score originality, relevance, feasibility, methodology, ethics compliance, and expected impact. Review outcomes: Approved, Minor Revision, Major Revision, Rejected. Complete review history must be stored.

7. One-Time M-Pesa Payment

Payment occurs only once during proposal submission. Uses STK Push. Stores CheckoutRequestID, MerchantRequestID, Receipt Number, Phone Number, Amount, Status, and Callback Data. Payment unlocks all subsequent stages without additional fees.

8. Research Approval & Clearance Certificate

Generated automatically after proposal approval. Includes Certificate Number, Research Code, Researcher Information, Institution, Study Sites, Validity Period, Committee Approval Statement, QR Verification Code, Signature Area, and Official Seal.

9. Stage 2 - Research Progress Submission

Purpose: Evaluate ongoing research before final publication. Required Information: Methodology, Study Design, Sampling Method, Sample Size Achieved, Data Collection Progress, Statistical Methods, Analysis Tools, Preliminary Findings, Deviations from Approved Protocol, Ethical Incidents, Participant Withdrawal Information. Uploads: Draft Manuscript, Datasets, Statistical Outputs, Survey Tools, Interview Guides, Images, Supporting Documents.

10. Progress Review Process

Reviewers assess: Methodology Compliance, Data Quality, Statistical Validity, Ethical Compliance, Research Progress. Recommendations: Continue, Minor Revision, Major Revision, Suspend Study.

11. Stage 3 - Final Paper Submission

Researchers submit: Final Manuscript, Publication Version, Final Dataset, Data Dictionary, Statistical Scripts, Consent Documentation, Ethics Approval Documents, Conflict of Interest Declarations, Funding Disclosures, AI Usage Declaration, Plagiarism Report.

12. Final Review Process

Reviewers validate scientific integrity, publication readiness, completeness of supporting documents, and compliance with institutional standards.

13. Publication & Repository Module

Store approved research papers. Features: DOI Storage, Publication Date, Journal Information, Download Statistics, Citation Information, Repository Search, Public Research Profiles.

14. Research Completion & Publication Certificate

Issued after final approval and publication. Contains Research Information, Publication Information, Completion Statement, Director Signature Area, Verification QR Code, and Certificate Number.

15. Notifications

Email and in-system notifications for: Payment Confirmation, Reviewer Assignment, Revision Requests, Approval Notices, Certificate Generation, Publication Approval.

16. MongoDB Collections

users researches researchreviews payments certificates notifications publications auditlogs

17. Research Document Structure

Research documents store stage information, payment status, reviewer assignments, certificates, publication data, and workflow history.

18. Audit Trail

Track every action: Submissions, Reviews, Status Changes, Payments, Certificate Generation, Downloads, Administrative Actions.

19. Dashboard Requirements

Researcher Dashboard: Active Projects, Status Tracking, Certificates, Reviewer Comments.
Reviewer Dashboard: Assigned Reviews, Pending Reviews, Review History. Admin Dashboard: Revenue Reports, Research Statistics, Review Assignments, Approval Metrics, Publication Analytics.

20. Security Requirements

JWT Authentication, Role-Based Access Control, File Validation, Secure Upload Storage, Audit Logging, Encryption of Sensitive Data.

21. Future Enhancements

NACOSTI Integration, KEMRI Workflow Templates, AI Research Assistant, Automated Plagiarism Checking, DOI Generation, Public Research Repository, Research Analytics Dashboard.